

MARY DAHL

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Education

- 2022 – 2025: **Candidate for a PhD in Aerospace Engineering**, MIT, Cambridge, MA.
(expected) Thesis: Autonomous and Dynamic Satellite Remote Sensing for Improved Cloud Observation
- 2020 – 2022: **M.S. in Aerospace Engineering**, MIT, Cambridge, MA.
Thesis: Development of Structures and Methods for Safe On Orbit Robotic Assembly of Small Satellites
- 2016 – 2020: **B.S. in Aerospace Engineering with a Minor in Creative Writing**, MIT, Cambridge, MA.

Research Experience

- Oct 2023 – present ***Dynamic Targeting and Intelligent Instrumentation for Cloud Science.***
- Developed mission concept for onboard decision making and autonomous pointing for efficient maximization of weather and climate data collection by defining requirements for dynamic measurements and analyzing mission architectures.
 - Analyzed various RF-based sensing systems, including radar and lidar, for their suitability for off-nadir and dynamic measurements.
- June 2024 – present ***SPECIES: Smart Polyimide Expandable Collector to enable Investigations for Earth Science.***
- Developed models for assessing the signal quality of a diffractive optical element as a lidar receiver payload on a small satellite for measuring cloud top height.
- May 2021 – May 2023 ***SEASALT: Satellite for Estimating Aquatic Salinity and Temperature.***
- Managed a multi-disciplinary team of engineers and students from MIT, Woods Hole Oceanographic Institution, and the Naval Research Laboratory to design a 6U satellite to measure ocean salinity through the use of a custom telescope optical payload and regression algorithms.
 - Led mission planning by developing requirements, design trades, technical budgets, interface documentation, and successfully leading two design reviews.
- Jan 2020 – May 2022 ***Modular CubeSat Assembly.***
- Designed 3D-printed, compliantly designed, snap fit shells for CubeSat components that could be manipulated and assembled by a cartesian robot safely and efficiently in less than 4 minutes.
 - Performed a Systems Theoretic Process Analysis (STPA) safety assessment on the robotic system to define potential anomalies for future investigations.

Industry Experience

- Feb 2023 – Aug 2023 ***MIT Lincoln Laboratory, Beaverworks Summer Institute Lead Instructor***, Cambridge, MA.
- Managed a class of 19 high school students for a full day summer program to teach them the basics of satellite engineering, Python, STK, and the use of amateur hardware such as Raspberry Pis, IMUs, radios, and solar panels.
- June 2019 – Aug 2019 ***SpaceX, Test Engineering Intern***, Hawthorne, CA.
- Developed and executed a large-scale pyrotechnics test as part of the Development Test Engineering Team to finalize the inclusion of a critical safety feature in the Dragon Crew Capsule.
 - Designed and fabricated custom parts, organized assembly and instrumentation of a test stand, planned and oversaw a large-scale lift operation, and coordinated with three teams of stakeholders.
- June 2018 – Aug 2018 ***MIT Lincoln Laboratory, Intern***, Lexington, MA.
- Designed and fabricated a test stand for verifying performance of experimental microbolometer IR cameras in different weather conditions for the Tactical Defense Systems Group.

Selected Publications and Presentations

- 2025 **Mary Dahl** and Kerri Cahoy. Improving Cloud Observations by Autonomously Pointing Satellites. *IEEE IGARSS 2025*. To be published.

- 2025 Adam Bahlous-Boldi et al. (including **Mary Dahl**). Deployable Diffractive Optical Elements for Small Satellite Lidar Missions. *39th Annual Small Satellite Conference*.
- 2025 **Mary Dahl**, Sophie Bergkvist, and Kerri Cahoy. Satellite System Design and Algorithms for Targeted Cloud Measurements. *NASA Earth and Planetary Cloud Workshop 2025*.
- 2023 **Mary Dahl**, Christine Page, Kerri Cahoy, and Evana Gizzi. Developing Intelligent Space Systems: a Survey and Rubric for Future Missions. in *37th Annual Small Satellite Conference*.
- 2022 Ezinne Uzo-Okoro, **Mary Dahl**, and Kerri Cahoy. Ground-based 1U CubeSat Robotic Assembly Demonstration. *Journal of Small Satellites, Volume 11*.
- 2022 Shreeyam Kacker et al. (including **Mary Dahl**). Satellite for Estimating Aquatic Salinity and Temperature (SEASALT) a Payload and Instrumentation Overview. *36th Annual Small Satellite Conference*.
- 2021 **Mary Dahl**, Juliana Chew, and Kerri Cahoy. Optimization of SmallSat Constellations and Low Cost Hardware to Utilize Onboard Planning. *ASCEND 2021*.

Honors

- May 23, 24 MathWorks Fellowship (awarded twice)
- May 2023 MIT AeroAstro Outstanding Graduate Student Leadership Recognition Award
- May 2022 Actor for *The V.E.R.S. Project*, 2022 Webby Award in Best Social Gaming Experience
- April 2022 MIT AeroAstro Community Recognition Graduate Teaching Assistantship Award

Leadership

- March 2023 – **Queer Advocacy Space in AeroAstro (QuASAR)**, *Co-Founder*, Cambridge, MA.
 - present
 - o Advocated for the inclusivity and empowerment of queer students in the Aerospace Engineering department by hosting teaching events, running social events, and performing community outreach.
- Jan 2021 – **Graduate Association of Aeronautics and Astronautics**, *Social Chair, Graduate Student Council Representative*, Cambridge, MA.
 - Jan 2024
 - o Planned and ran social events for approximately 300 graduate students, including off-campus trips, office trick-or-treating, coffee socials, and new student mixers.
 - o Worked with the interdisciplinary Graduate Student Council to advocate for the needs of AeroAstro students during the transition back to in-person instruction and activities.
- Jan 2017 – **MIT Shakespeare Ensemble**, *Technical Director, Producer, Actor*, Cambridge, MA.
 - May 2020
 - o Recruited and lead teams of approximately 20 students to put together productions of theater shows by managing requirements, budgets, deadlines, and conflicts.
 - o Coordinated the full-day technical setup by managing students across multiple teams to complete set construction, lighting and sound setup, audience arrangements, and backstage preparations.
- April 2017 – **MIT Next House Dormitory Exec**, *Vice President, Social Chair*, Cambridge, MA.
 - Dec 2019
 - o Managed student needs by working with an executive board, such as by building infrastructure and securing funds for organized student group activities and running housewide meetings.
- Sep 2016 – **MIT Next Haunt**, *Producer, Assistant Producer, Build Director*, Cambridge, MA.
 - Nov 2019
 - o Lead a team of approximately 15 students to design and construct a wooden, two-story haunted house escape room, featuring crawl tunnels, structures for hiding actors, and electronic puzzles.
 - o Conducted operation of event for multiple weekends by coordinating “zombie” actors, properly timing hint delivery for optimal experiences, and rapidly responding to anomalies.

Professional Development

- 2023 NASA/CCS/KISS Summer School, “Using Satellite Observations to Advance Climate Models”
- 2019 Gordon-MIT Engineering Leadership Program

Skills

- Software Python, MATLAB, Solidworks, Ansys Mechanical, Ansys STK, Linux, Adobe Suite, L^AT_EX
- Machining Laser cutting, 3D printing, machine shop fabrication, power tools